MADE BRANENDA JORDHY

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EDUCATION

Bandung Institute of Technology Aug 2024 – present
B.E. in Informatics Engineering, GPA: 3.79 / 4.0 *Bandung, West Java*

- **Relevant Coursework:** Operating System, Software Engineering, Geometric and Linear Algebra, Artificial Intelligence and Data Literacy, Computer Organization and Architecture, Algorithm and Programming, Computational Logic, Discrete Mathematics
- **Member:** HMIF ITB, IEEE ITB Student Branch

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Go, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Chi, sqlc, TailwindCSS, Prisma
Databases: PostgreSQL, CockroachDB, Supabase, Cloudflare D1/R2
DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Cloudflare, AWS, Vercel, Azure

AWARDS

Top 20 at AI Innovation Challenge, COMPFEST 18 (University of Indonesia) | RAG, pgvector, PyTorch, React | GitHub 2026

- Built **SIENA**, an evidence-grounded **predictive-maintenance** platform for SME factories that fuses sensor anomaly detection, a deterministic machine health score, and **pgvector RAG** over SOPs and maintenance history into LLM-reasoned, human-approved work orders. Fine-tuned a **MobileNetV3** QC defect classifier (MVTec AD, class-balanced sampling) mapped to machine failure modes, and built the React + TypeScript frontend end to end.

Semifinalist at AI Open Innovation Challenge, President University | XGBoost, YOLOv8, MapLibre | GitHub July 2026

- Built **JakaLink**, an AI orchestration layer for DISHUB DKI Jakarta unifying TransJakarta, MRT, LRT, and KRL into one network, with **conformal-interval ETAs** and **crowding forecasts**. Built the Apple Maps-style commuter PWA (React, TypeScript, MapLibre GL) with multimodal trip planning, live route progress, and mid-journey rerouting.

2nd Place at Hackathon, Hackfest 2026 (Ciputra University) | DeepForest, NDVI, FastAPI, Next.js | GitHub April 2026

- Built **CarbonLink**, a digital **MRV carbon-credit** platform for Indonesian smallholder agroforestry, running **DeepForest** tree detection on aerial imagery and fusing it with NDVI and polygon land area to estimate CO_2e (IPCC Tier 1) through a 5-stage verification flow.

CERTIFICATIONS

Introduction to Back-End Development | Meta | Credential Issued Feb 2026
Programming in Python | Meta | Credential Issued Feb 2026

EXPERIENCES

STEI ITB Bandung, West Java
Software Engineering Laboratory Assistant *Aug 2026 – present*

- Assist **IF2150 Software Engineering**, guiding student teams through the full **SDLC**, from project ideation, **requirements elicitation**, and **user stories** to construction and deployment, across **Waterfall, Iterative, Scrum, and Agile** process models.
- Mentor **analysis and design** with **UML** modeling (**activity, swimlane, use case, sequence, and class diagrams**), covering **design concepts, strategies, and architectural design**, then reviewing artifacts for consistency against their implementation.
- Coach **verification & validation**, from **requirements validation** and **design evaluation** to **structured and automated testing** of the delivered build.
- Teach **software configuration management** and **Git best practices** (branching, code review, collaborative workflows), guiding students to organize engineering work as a **team** under **professional ethics**.

Inkubator IT (IIT) Bandung, West Java
DevOps Engineer (Technology Development) *May 2026 – present*

- Built and maintained **CI/CD pipelines** with **GitHub Actions and Docker** at a software house, automating build and deployment across multiple client web and mobile products with reliable, repeatable production releases.
- Deployed and operated applications on the **Cloudflare** edge stack (**Workers, Pages, D1, R2 object storage**) for fast global delivery, and wired **AWS SES** for transactional email (account verification, password resets, notifications).
- Evaluating a **K3s (lightweight Kubernetes)** setup to share infrastructure across client apps, letting each engagement start small and **scale horizontally** on demand.

Concorde Systems Bandung, West Java
Software Engineer *May 2026 – present*

- Owning the backend end to end at a **startup** building **ConcordeOS**, an AI-powered **offline-first** maintenance platform bridging enterprise CMMS (MaintainX) and field technicians in extreme industrial environments.
- Designed the **database schema (ERD)** and **REST API** from scratch, implementing type-safe endpoints with **Go (Chi, sqlc)** over **Supabase (PostgreSQL)**, plus JWT auth middleware and realtime data flows.
- Architecting an **asynchronous offline-first sync** layer that guarantees zero data loss in remote and underground sites, cutting MTTR for F&B, heavy manufacturing, and oil & gas clients.

Leadership & Activities

The Sandbox 3.0 Bandung, West Java
Director of Technology *Nov 2025 – May 2026*

- Led end-to-end engineering of a competition platform for **400+ users**, shipping the full registration and multi-stage submission pipeline (preliminary to final) with **Next.js, Prisma, and CockroachDB**.
- Built an internal admin panel for real-time financial verification and submission management, cutting organizers' manual review effort across the event cycle.

Arkavidia Bandung, West Java
Software Engineering Curriculum Developer *May 2026 – present*

- Designed a hands-on **software engineering curriculum** covering backend architecture, API integration, and production deployment for mentees.
- Led milestone-based build sessions on real features, guiding mentees to ship deployable end-to-end projects.

TEDxITB 9.0 Bandung, West Java
Full-Stack Developer *Nov 2025 – May 2026*

- Shipped full-stack features for the official **TEDxITB website**, building user flows from account registration to content discovery across server and client.
- Delivered API integrations, database logic, and responsive UI for a **high-traffic public platform**, collaborating via GitHub-based workflows.

IEEE ITB Student Branch Batch 2025/2026 Bandung, West Java
Full-Stack Developer *Aug 2025 – June 2026*

- Engineered the **IEEE Paper Explorer**, a data pipeline that fetches, processes, and renders live research-paper data from the **IEEE API** with dynamic content delivery.

Projects

Selected work (See all)

Source, please? | React, TypeScript, Cloudflare Durable Objects, Supabase | UNESCO Youth Hackathon 2026 | GitHub Aug 2026

- Built a **real-time multiplayer media-literacy game** where players back claims with evidence cards; wording a claim stronger than the evidence allows (*overclaim*) earns reach but costs credibility and is publicly corrected the next round.
- Engineered a **server-clocked round engine** on **Cloudflare Durable Objects** (PartyServer WebSockets) with readiness checks and practice bots, so single player runs on the same engine as multiplayer; wired Google sign-in, **Supabase RLS**, and persisted EXP and leaderboards.

domTraverse | Next.js, React, Go, Docker, Azure | GitHub May 2026

- Built a **DOM traversal visualizer** that scrapes any target URL, parses raw markup into a real tree, and animates **BFS and DFS** walks step by step with a full visit/match log.
- Implemented a **CSS-selector engine** (tag, class, id, and combinators) and real-time **Lowest Common Ancestor** computation; shipped with a **Go** backend (Next.js 16, React 19, TailwindCSS), Dockerized on **Microsoft Azure**.

VeilGuard | TypeScript, Chromium Extension API | GitHub May 2026

- Built a **Chromium browser extension** that detects online gambling (judol) content on web pages in real time, scanning the live DOM and flagging matches automatically as pages load.
- Implemented exact **string-matching** from scratch with the **Knuth-Morris-Pratt** (failure function, O(n+m)) and **Boyer-Moore** (bad-character heuristic) algorithms, packaged as a production bundle (Node.js, pnpm).

Eigen Pustaka | Next.js (TypeScript), FastAPI (Python) | GitHub Nov 2025

- Built a **book discovery engine** with TF-IDF + LSA text search, PCA image search, and personalized recommendations — all linear algebra hand-implemented without external ML libraries.
- Shipped a full-stack app (**Next.js + TypeScript, FastAPI + Python**) serving semantic text retrieval, image similarity, and recommendation workflows.

YAREU | Flet (Python), PostgreSQL, MVC Architecture | GitHub Nov 2025

- Developed a **sustainability platform** unifying buy/sell, donations, and trade-in flows in a single user experience.
- Implemented modular Marketplace, Donations, and Trade-In subsystems with **MVC architecture** (Flet, Python, PostgreSQL).

Research Papers

Dynamic Programming for Battery Energy Storage Arbitrage Scheduling Considering Capacity Degradation Cost June 2026
STEI ITB | Paper

- Topics: Dynamic Programming, Energy Storage, Battery Arbitrage, Degradation Cost, Optimal Scheduling, Algorithm Strategy.

Efficient Transformer Compression in Pre-trained Language Models Through Selective Tensor Rank Reduction Dec 2025
STEI ITB | Paper

- Topics: Model Compression, Natural Language Processing, Transformer Efficiency, Tensor Decomposition, Low Rank Approximation, Geometric and Linear Algebra.

Anomaly Detection in AI Knowledge Graphs: A Discrete Mathematical Framework Against Graph Poisoning June 2025
STEI ITB | Paper

- Topics: Knowledge Graphs, AI Security, Discrete Mathematics, Graph Theory, Anomaly Detection.